

HR ESS Portal

Schema Design & Module Architecture

Stack	Java 21 + Spring Boot 3 + Vue 3 + PostgreSQL
Auth	SAML 2.0 (Azure AD / Okta) → JWT
Storage	AWS S3 — payslips, receipts, employee docs
Email	AWS SES — templated notifications
Realtime	Server-Sent Events (SSE)
Batch	Spring Batch — leave accrual, payslip publish
Deploy	Docker → AWS ECR → ECS Fargate via GitHub Actions
DB Tables	29 tables across 8 modules
Backend Modules	10 packages — auth, employee, leave, expense, payroll, workflow, notification, reports, admin, common

Part 1 — Database Schema

PostgreSQL 15+ · UUID primary keys · TIMESTAMP columns use timezone=UTC · All tables have created_at DEFAULT now(); sensitive columns encrypted via pgcrypto

Authentication & Users

3 tables

roles

Column	Type	Constraint	Description
id	BIGINT	PK	Auto-increment primary key
role_name	VARCHAR(50)	UNIQUE	EMPLOYEE, MANAGER, HR_ADMIN, FINANCE_ADMIN, SYSTEM_A
description	TEXT	NOT NULL	Human readable role description
Static seed data — 5 roles; do not allow runtime inserts			

users

Column	Type	Constraint	Description
id	UUID	PK	gen_random_uuid()
employee_id	UUID	FK → employees.id	Nullable for system users
email	VARCHAR(255)	UNIQUE NOT NULL	Login email — matches SAML NameID
role_id	BIGINT	FK → roles.id	Single role per user
status	VARCHAR(20)	NOT NULL	ACTIVE, INACTIVE, LOCKED
last_login	TIMESTAMP	DEFAULT	Updated on each login
created_at	TIMESTAMP	DEFAULT	now()
One user per employee; role drives all access control decisions			

refresh_tokens

Column	Type	Constraint	Description
id	UUID	PK	Token ID
user_id	UUID	FK → users.id NOT NULL	Owning user
token_hash	VARCHAR(255)	UNIQUE NOT NULL	SHA-256 hash of token
expires_at	TIMESTAMP	NOT NULL	30 day expiry
revoked	BOOLEAN	DEFAULT	false — set true on logout
JWT refresh tokens; pruned by scheduler every 24h			

Table Relationships

users	many → 1	roles	Each user has exactly one role
users	1 → 1	employees	Each user links to one employee record
users	1 → many	refresh_tokens	User can have multiple active tokens

Employee & Onboarding

5 tables

employees

Column	Type	Constraint	Description
id	UUID	PK	gen_random_uuid()
employee_code	VARCHAR(20)	UNIQUE NOT NULL	e.g. EMP-2024-0042
full_name	VARCHAR(200)	NOT NULL	Display name
email	VARCHAR(255)	UNIQUE NOT NULL	Corporate email
phone	VARCHAR(20)		Mobile number
department	VARCHAR(100)	NOT NULL	Engineering, HR, Finance...
designation	VARCHAR(100)	NOT NULL	Job title
employment_type	VARCHAR(20)	NOT NULL	FULL_TIME, PART_TIME, CONTRACT
manager_id	UUID	FK → employees.id	Self-referencing — nullable for CEO
joining_date	DATE	NOT NULL	Contract start date
status	VARCHAR(20)	NOT NULL	ACTIVE, INACTIVE, ON_NOTICE
profile_photo	VARCHAR(500)		S3 URL
created_at	TIMESTAMP	DEFAULT	now()

Core employee master — source of truth for all HR operations

employee_bank_details

Column	Type	Constraint	Description
id	UUID	PK	
employee_id	UUID	FK → employees.id	
bank_name	VARCHAR(100)	NOT NULL	
account_no	BYTEA	NOT NULL	pgcrypto encrypted
ifsc_code	VARCHAR(20)	NOT NULL	
account_type	VARCHAR(20)		SAVINGS, CURRENT
is_active	BOOLEAN	DEFAULT	true — only one active at a time

Encrypted at rest via PostgreSQL pgcrypto; only Finance Admin can decrypt

employee_contacts

Column	Type	Constraint	Description
id	UUID	PK	
employee_id	UUID	FK → employees.id	
name	VARCHAR(200)	NOT NULL	Contact full name
relationship	VARCHAR(50)	NOT NULL	Spouse, Parent, Sibling...
phone	VARCHAR(20)	NOT NULL	
email	VARCHAR(255)		Optional

Emergency contacts — minimum one required before onboarding completion

employee_documents			
Column	Type	Constraint	Description
id	UUID	PK	
employee_id	UUID	FK → employees.id	
doc_type	VARCHAR(50)	NOT NULL	OFFER_LETTER, ID_PROOF, CERTIFICATE...
file_name	VARCHAR(255)	NOT NULL	Original filename
s3_key	VARCHAR(500)	NOT NULL	S3 object key
s3_bucket	VARCHAR(100)	NOT NULL	Bucket name
expiry_date	DATE		Nullable — for visa/cert expiry
uploaded_by	UUID	FK → users.id	Who uploaded
All HR documents linked to an employee; versioned via created_at			

onboarding_tasks			
Column	Type	Constraint	Description
id	UUID	PK	
employee_id	UUID	FK → employees.id	
task_name	VARCHAR(200)	NOT NULL	e.g. Upload ID proof
task_type	VARCHAR(50)	NOT NULL	DOCUMENT_UPLOAD, FORM_FILL, ACKNOWLEDGEMENT
status	VARCHAR(20)	NOT NULL	PENDING, COMPLETED
completed_at	TIMESTAMP		Nullable
due_date	DATE		Optional deadline
Template tasks assigned to each new employee on creation			

Table Relationships			
employees	1 → many	employee_documents	All employee files
employees	1 → many	employee_contacts	Emergency contacts
employees	1 → 1	employee_bank_details	Bank for payroll
employees	1 → many	onboarding_tasks	Onboarding checklist
employees	many → 1	employees	Self-ref manager hierarchy

Leave Management

5 tables

leave_policies

Column	Type	Constraint	Description
id	UUID	PK	
leave_type	VARCHAR(50)	UNIQUE NOT NULL	ANNUAL, CASUAL, SICK, MATERNITY, PATERNITY
accrual_frequency	VARCHAR(20)	NOT NULL	MONTHLY, QUARTERLY, ANNUAL
accrual_days	DECIMAL(5,2)	NOT NULL	Days accrued per period
max_balance	DECIMAL(5,2)	NOT NULL	Max carry-forward cap
carry_forward	BOOLEAN	DEFAULT	false
carry_forward_max	DECIMAL(5,2)		Max days to carry forward
applicable_to	VARCHAR(20)	NOT NULL	ALL, FEMALE, MALE
notice_days	INT	DEFAULT	Minimum advance notice
Defines rules per leave type — configured by HR Admin			

leave_balances

Column	Type	Constraint	Description
id	UUID	PK	
employee_id	UUID	FK → employees.id	
leave_type	VARCHAR(50)	NOT NULL	FK concept — matches leave_policies.leave_type
year	INT	NOT NULL	Calendar year
total_days	DECIMAL(5,2)	NOT NULL	Accrued total
used_days	DECIMAL(5,2)	NOT NULL	Approved and taken
pending_days	DECIMAL(5,2)	NOT NULL	Pending approval
balance	DECIMAL(5,2)	NOT NULL	total - used - pending
Updated monthly by Spring Batch accrual job			

leave_requests

Column	Type	Constraint	Description
id	UUID	PK	
employee_id	UUID	FK → employees.id	
leave_type	VARCHAR(50)	NOT NULL	
start_date	DATE	NOT NULL	
end_date	DATE	NOT NULL	
total_days	DECIMAL(5,2)	NOT NULL	Computed excluding holidays
reason	TEXT	NOT NULL	Employee's note
status	VARCHAR(20)	NOT NULL	PENDING, APPROVED, REJECTED, CANCELLED
workflow_id	UUID	FK → workflow_requests.id	links to workflow engine
applied_at	TIMESTAMP	DEFAULT	
Core leave application — drives the approval workflow			

holiday_calendar			
Column	Type	Constraint	Description
id	UUID	PK	
date	DATE	NOT NULL	Holiday date
name	VARCHAR(200)	NOT NULL	e.g. Diwali, Christmas
country	VARCHAR(5)	NOT NULL	IN, US, UK
is_optional	BOOLEAN	DEFAULT	false — optional holidays
Shared by leave calculation logic and calendar UI			

leave_accrual_logs			
Column	Type	Constraint	Description
id	UUID	PK	
employee_id	UUID	FK → employees.id	
leave_type	VARCHAR(50)	NOT NULL	
days_accrued	DECIMAL(5,2)	NOT NULL	
accrual_date	DATE	NOT NULL	Batch run date
reason	VARCHAR(100)	NOT NULL	MONTHLY_ACCRUAL, CARRY_FORWARD, ADJUSTMENT
Immutable audit log written by Spring Batch — never updated			

Table Relationships			
leave_requests	many → 1	employees	Request belongs to employee
leave_requests	1 → 1	workflow_requests	Approval via workflow engine
leave_balances	many → 1	employees	One balance record per leave type per year
leave_accrual_logs	many → 1	employees	Monthly accrual audit

Expense Claims

4 tables

expense_claims

Column	Type	Constraint	Description
id	UUID	PK	
employee_id	UUID	FK → employees.id	
category	VARCHAR(50)	NOT NULL	TRAVEL, MEALS, ACCOMMODATION, OTHER
amount	DECIMAL(12, 2)	NOT NULL	Amount in original currency
currency	VARCHAR(3)	NOT NULL	ISO 4217 e.g. INR, USD
base_amount	DECIMAL(12, 2)	NOT NULL	Normalised to INR using rate at submission
exchange_rate	DECIMAL(10, 4)	NOT NULL	Rate applied at submission time
description	TEXT	NOT NULL	Employee's explanation
expense_date	DATE	NOT NULL	Date of expenditure
status	VARCHAR(30)	NOT NULL	DRAFT, PENDING, MGR_APPROVED, FIN_APPROVED, PAID, RE
workflow_id	UUID	FK → workflow_requests.id	
payment_ref	VARCHAR(100)		Set by Finance on payment
paid_at	TIMESTAMP		Nullable
created_at	TIMESTAMP	DEFAULT	
Master claim record; status updated by workflow events			

expense_receipts

Column	Type	Constraint	Description
id	UUID	PK	
expense_id	UUID	FK → expense_claims.id	
file_name	VARCHAR(255)	NOT NULL	Original filename
s3_key	VARCHAR(500)	NOT NULL	S3 object key
file_type	VARCHAR(20)	NOT NULL	IMAGE, PDF
file_size_kb	INT	NOT NULL	For storage reporting
Multiple receipts per claim; stored in S3			

expense_limits

Column	Type	Constraint	Description
id	UUID	PK	
category	VARCHAR(50)	NOT NULL	
limit_type	VARCHAR(20)	NOT NULL	PER_DAY, PER_TRIP, PER_MONTH
max_amount	DECIMAL(12, 2)	NOT NULL	In base currency (INR)
applies_to	VARCHAR(50)	NOT NULL	ALL or specific grade/department
is_active	BOOLEAN	DEFAULT	true
Policy limits enforced at submission; configurable by HR Admin			

currency_rates			
Column	Type	Constraint	Description
id	UUID	PK	
from_currency	VARCHAR(3)	NOT NULL	Source currency code
to_currency	VARCHAR(3)	NOT NULL	INR default base
rate	DECIMAL(10, 4)	NOT NULL	1 unit of from = rate units of to
effective_date	DATE	NOT NULL	Date this rate is valid from
created_by	UUID	FK → users.id	
Manually updated by Finance Admin; used at claim submission time			

Table Relationships			
expense_claims	many → 1	employees	Claim belongs to employee
expense_claims	1 → many	expense_receipts	One claim, multiple receipts
expense_claims	1 → 1	workflow_requests	Approval via workflow engine
expense_limits	policy for	expense_claims	Enforced at submission

Payroll & Payslips

3 tables

payroll_uploads

Column	Type	Constraint	Description
id	UUID	PK	
uploaded_by	UUID	FK → users.id	HR Admin who uploaded
file_name	VARCHAR(255)	NOT NULL	Original CSV filename
s3_key	VARCHAR(500)	NOT NULL	S3 key of uploaded CSV
month	INT	NOT NULL	1-12
year	INT	NOT NULL	
total_rows	INT	NOT NULL	Rows in CSV
valid_rows	INT	NOT NULL	
error_rows	INT	NOT NULL	
status	VARCHAR(20)	NOT NULL	UPLOADED, VALIDATING, VALID, INVALID, GENERATING, RE
published_at	TIMESTAMP		Set when HR clicks Publish
Each CSV upload is one batch; validated before payslips are generated			

payroll_errors

Column	Type	Constraint	Description
id	UUID	PK	
upload_id	UUID	FK → payroll_uploads.id	
row_number	INT	NOT NULL	CSV row with error
employee_code	VARCHAR(20)	NOT NULL	From CSV
error_type	VARCHAR(100)	NOT NULL	UNKNOWN_EMPLOYEE, MISSING_FIELD, NEGATIVE_AMOUNT...
Row-level errors from CSV validation — shown in upload review screen			

payslips			
Column	Type	Constraint	Description
id	UUID	PK	
employee_id	UUID	FK → employees.id	
upload_id	UUID	FK → payroll_uploads.id	Source batch
month	INT	NOT NULL	1-12
year	INT	NOT NULL	
basic_pay	DECIMAL(12,2)	NOT NULL	
hra	DECIMAL(12,2)	NOT NULL	
other_allowances	DECIMAL(12,2)	NOT NULL	
gross_pay	DECIMAL(12,2)	NOT NULL	Sum of earnings
pf_deduction	DECIMAL(12,2)	NOT NULL	Provident Fund
tax_deduction	DECIMAL(12,2)	NOT NULL	TDS
other_deductions	DECIMAL(12,2)	NOT NULL	
total_deductions	DECIMAL(12,2)	NOT NULL	
net_pay	DECIMAL(12,2)	NOT NULL	gross - deductions
pdf_s3_key	VARCHAR(500)	NOT NULL	Password-protected PDF
published	BOOLEAN	DEFAULT	false — visible to employee when true
published_at	TIMESTAMP		
One row per employee per month; PDF generated and stored in S3			

Table Relationships			
payroll_uploads	1 → many	payslips	One batch generates many payslips
payroll_uploads	1 → many	payroll_errors	Validation errors per upload
payslips	many → 1	employees	Employee's monthly payslip

Workflow Engine

3 tables

workflow_requests

Column	Type	Constraint	Description
id	UUID	PK	
entity_type	VARCHAR(50)	NOT NULL	LEAVE, EXPENSE, PROFILE_CHANGE
entity_id	UUID	NOT NULL	FK to respective module's table
requester_id	UUID	FK → users.id	Employee who initiated
current_step	INT	NOT NULL	Current step in workflow (1, 2, ...)
total_steps	INT	NOT NULL	Total steps for this entity type
status	VARCHAR(20)	NOT NULL	PENDING, APPROVED, REJECTED, CANCELLED
created_at	TIMESTAMP	DEFAULT	
Central workflow table — reused by leave, expense, and future modules			

workflow_actions

Column	Type	Constraint	Description
id	UUID	PK	
workflow_id	UUID	FK → workflow_requests.id	
step	INT	NOT NULL	Step number this action belongs to
action_by	UUID	FK → users.id	Who took action
action	VARCHAR(20)	NOT NULL	SUBMITTED, APPROVED, REJECTED, COMMENT, CANCELLED
comment	TEXT		Mandatory on REJECTED
Immutable log of every approval, rejection, or comment — never deleted			

workflow_delegates

Column	Type	Constraint	Description
id	UUID	PK	
delegator_id	UUID	FK → users.id	Manager going on leave
delegate_id	UUID	FK → users.id	Acting manager
start_date	DATE	NOT NULL	
end_date	DATE	NOT NULL	
is_active	BOOLEAN	DEFAULT	true
Manager delegation during leave — approval authority transfer			

Table Relationships

workflow_requests	1 → many	workflow_actions	Full audit trail per request
workflow_requests	entity_id →	leave_requests / expense_claims	Polymorphic entity ref
workflow_delegates	many → 1	users	Delegator and delegate both FK to users

Notifications

3 tables

notifications

Column	Type	Constraint	Description
id	UUID	PK	
user_id	UUID	FK → users.id	Recipient
event_type	VARCHAR(50)	NOT NULL	LEAVE_APPROVED, EXPENSE_PAID, PAYSLIP_PUBLISHED...
title	VARCHAR(200)	NOT NULL	Short notification title
message	TEXT	NOT NULL	Full notification body
entity_type	VARCHAR(50)		LEAVE, EXPENSE, PAYSLIP
entity_id	UUID		For deep-link navigation
is_read	BOOLEAN	DEFAULT	false
read_at	TIMESTAMP		Nullable
In-app notifications delivered via SSE; email sent via AWS SES			

notification_preferences

Column	Type	Constraint	Description
id	UUID	PK	
user_id	UUID	FK → users.id UNIQUE	One row per user
event_type	VARCHAR(50)	NOT NULL	
email_enabled	BOOLEAN	DEFAULT	true
inapp_enabled	BOOLEAN	DEFAULT	true
Per-user preferences; HR Admin can set org-wide defaults			

email_templates

Column	Type	Constraint	Description
id	UUID	PK	
event_type	VARCHAR(50)	UNIQUE NOT NULL	
subject	VARCHAR(500)	NOT NULL	Supports {{employee_name}} etc.
body_html	TEXT	NOT NULL	Full HTML email template
updated_by	UUID	FK → users.id	
Editable via Admin UI; supports Handlebars-style placeholders			

Table Relationships

notifications	many → 1	users	Recipient user
notification_preferences	many → 1	users	Preference per user
email_templates	used by	notifications service	Rendered on event dispatch

Admin & Config

3 tables

system_settings

Column	Type	Constraint	Description
id	UUID	PK	
setting_key	VARCHAR(100)	UNIQUE NOT NULL	e.g. SESSION_TIMEOUT_MINUTES
setting_value	TEXT	NOT NULL	Stored as string
description	TEXT		Admin-facing description
updated_by	UUID	FK → users.id	

Key-value config store; all values are strings — cast in application layer

audit_logs

Column	Type	Constraint	Description
id	UUID	PK	
user_id	UUID	FK → users.id	Who performed action
action	VARCHAR(100)	NOT NULL	CREATE, UPDATE, DELETE, LOGIN, LOGOUT
entity_type	VARCHAR(50)	NOT NULL	Table / module name
entity_id	UUID		Nullable for non-record actions
old_value	JSONB		Before state
new_value	JSONB		After state
ip_address	VARCHAR(45)	NOT NULL	IPv4 or IPv6
user_agent	TEXT		Browser/app string

Immutable compliance log — insert-only, never updated or deleted

scheduler_jobs

Column	Type	Constraint	Description
id	UUID	PK	
job_name	VARCHAR(100)	UNIQUE NOT NULL	e.g. MONTHLY_LEAVE_ACCRUAL
cron_expr	VARCHAR(50)	NOT NULL	Standard cron: 0 0 1 * *
is_enabled	BOOLEAN	DEFAULT	true
last_run_at	TIMESTAMP		
last_run_status	VARCHAR(20)		SUCCESS, FAILED, RUNNING

Spring Batch job registry with manual trigger capability

Table Relationships

audit_logs	many → 1	users	Every log entry has an actor
system_settings	read by	all modules	Global config source
scheduler_jobs	run by	Spring Batch	Job execution framework

Part 2 — Backend Module Design

Package structure: `com.company.hrportal.{module}` · Each module owns its Controller → Service → Repository layers · Cross-module communication via Spring `ApplicationEvents`, never direct service calls

`com.company.hrportal.auth`

Authentication, JWT, SAML, Spring Security config

Class	Type	Responsibility
<code>SamlAuthController</code>	Controller	Handles SAML assertion consumer, redirects
<code>JwtService</code>	Service	Generate, validate, refresh JWT tokens
<code>AuthService</code>	Service	Login, logout, session management
<code>JwtAuthFilter</code>	Filter	Extends <code>OncePerRequestFilter</code> — validates JWT
<code>SecurityConfig</code>	Config	Spring Security filter chain, CORS, CSRF
<code>UserPrincipal</code>	Model	Implements <code>UserDetails</code> — loaded per request

API Endpoints

POST	<code>/api/auth/saml/acs</code>	SAML Assertion Consumer Service
POST	<code>/api/auth/refresh</code>	Refresh JWT using refresh token
POST	<code>/api/auth/logout</code>	Invalidate refresh token
GET	<code>/api/auth/me</code>	Returns current user's profile + role

com.company.hrportal.employee

Employee profile, onboarding, documents, directory

Class	Type	Responsibility
EmployeeController	Controller	Profile CRUD, directory, onboarding APIs
EmployeeService	Service	Business logic, onboarding task auto-create
EmployeeRepository	Repository	JPA — employees, contacts, bank, docs tables
OnboardingService	Service	Task management, completion tracking
DocumentService	Service	S3 upload/download via FileService
EmployeeMapper	Mapper	Entity ↔ DTO using MapStruct

API Endpoints

GET	/api/employees	List/search (HR Admin, Manager)
POST	/api/employees	Create new employee (HR Admin)
GET	/api/employees/{id}	Get employee detail
PUT	/api/employees/{id}	Update profile
GET	/api/employees/{id}/onboarding	Onboarding tasks
PUT	/api/onboarding/{taskId}/complete	Mark task done
POST	/api/employees/{id}/documents	Upload document to S3

com.company.hrportal.leave

Leave requests, balances, policy, calendar, accrual batch

Class	Type	Responsibility
LeaveController	Controller	Apply, cancel, approve, reject APIs
LeaveService	Service	Submit → workflow, balance deduction logic
LeaveBalanceService	Service	Balance calc, accrual, carry-forward
LeavePolicyService	Service	Policy CRUD for HR Admin
LeaveRepository	Repository	leave_requests, balances, policies, logs
HolidayCalendarService	Service	Holiday lookup for working day calculation
LeaveAccrualJob	Batch	Spring Batch — monthly accrual step
CarryForwardJob	Batch	Spring Batch — year-end carry forward

API Endpoints

GET	/api/leave/balance	My leave balance per type
POST	/api/leave/apply	Submit leave request
PUT	/api/leave/{id}/cancel	Cancel pending request
GET	/api/leave/calendar	Calendar view with team + holidays
GET	/api/manager/leave/pending	Pending approvals (Manager)
PUT	/api/manager/leave/{id}/approve	Approve with comment
PUT	/api/manager/leave/{id}/reject	Reject with reason
GET	/api/admin/leave-policy	List policies (HR Admin)
PUT	/api/admin/leave-policy/{id}	Update policy

com.company.hrportal.expense

Expense claims, receipts, multi-currency, finance queue

Class	Type	Responsibility
ExpenseController	Controller	Submit, track, manager and finance APIs
ExpenseService	Service	Submit → workflow, limit validation, currency
ExpenseLimitService	Service	Policy limit check at submission
CurrencyService	Service	Rate lookup and amount normalisation
ReceiptService	Service	Receipt upload to S3, pre-signed URL
ExpenseRepository	Repository	expense_claims, receipts, limits, rates
FinanceExpenseService	Service	Finance queue — approve, pay, export

API Endpoints

POST	/api/expense	Submit new claim
GET	/api/expense	My claims list
PUT	/api/expense/{id}/cancel	Cancel draft/pending
GET	/api/manager/expense/pending	Team pending approvals
PUT	/api/manager/expense/{id}/approve	
PUT	/api/manager/expense/{id}/reject	
GET	/api/finance/expense/queue	Finance approval queue
PUT	/api/finance/expense/{id}/pay	Mark paid with reference

com.company.hrportal.payroll

CSV upload, validation, payslip generation, PDF, publish

Class	Type	Responsibility
PayrollController	Controller	Upload, validate, preview, publish APIs
PayrollUploadService	Service	CSV parsing, validation, error collection
PayslipGeneratorService	Service	Payslip row → PDF generation via iText
PayslipPdfService	Service	Password-protect PDF, upload to S3
PayrollPublishService	Service	Batch publish — trigger notifications
PayrollRepository	Repository	payroll_uploads, payslips, errors
PayslipPublishJob	Batch	Spring Batch — async publish step

API Endpoints

POST	/api/hr/payroll/upload	Upload CSV (multipart)
GET	/api/hr/payroll/{uploadId}/errors	Validation errors
POST	/api/hr/payroll/{uploadId}/generate	Trigger generation
POST	/api/hr/payroll/{uploadId}/publish	Publish batch
GET	/api/payslip	My payslips list
GET	/api/payslip/{id}/download	Download PDF (pre-signed S3 URL)

com.company.hrportal.workflow

Reusable approval engine — used by leave, expense, future modules

Class	Type	Responsibility
WorkflowService	Service	Create, advance, resolve workflow requests
WorkflowActionService	Service	Record approve/reject/comment actions
WorkflowRepository	Repository	workflow_requests, actions, delegates
WorkflowEventPublisher	Service	Publishes Spring ApplicationEvents on status change
DelegationService	Service	Manager delegation CRUD + routing logic
WorkflowConfig	Config	Step definitions per entity type

API Endpoints

GET	/api/workflow/{id}	Workflow status + history
GET	/api/workflow/my-pending	All my pending action items
POST	/api/admin/delegation	Set up manager delegation
DELETE	/api/admin/delegation/{id}	Remove delegation

com.company.hrportal.notification

SSE real-time, AWS SES email, in-app notifications

Class	Type	Responsibility
NotificationController	Controller	SSE endpoint, list, mark-read APIs
NotificationService	Service	Create notification, dispatch email + in-app
SseService	Service	SseEmitter management, broadcast per user
EmailService	Service	AWS SES client, template rendering
TemplateRenderer	Service	Handlebars template merge with event data
WorkflowEventListener	Listener	@EventListener — reacts to workflow events
NotificationRepository	Repository	notifications, preferences, templates

API Endpoints

GET	/api/notifications/stream	SSE endpoint (text/event-stream)
GET	/api/notifications	Paginated notification list
PUT	/api/notifications/{id}/read	Mark one as read
PUT	/api/notifications/read-all	Mark all as read
GET	/api/notifications/preferences	My preferences
PUT	/api/notifications/preferences	Update preferences

com.company.hrportal.reports

Headcount, leave utilisation, expense summary — CSV/Excel export

Class	Type	Responsibility
ReportController	Controller	Headcount, leave, expense report APIs
HeadcountReportService	Service	Employee count aggregations by filters
LeaveReportService	Service	Leave utilisation queries
ExpenseReportService	Service	Expense summary with category breakdown
ExcelExportService	Service	Apache POI — generate .xlsx
CsvExportService	Service	OpenCSV — generate .csv

API Endpoints

GET	/api/reports/headcount	Headcount report (filters: dept, type, date)
GET	/api/reports/leave	Leave utilisation report
GET	/api/reports/expense	Expense summary report
GET	/api/reports/headcount/export?format=csv xlsx	
GET	/api/reports/leave/export?format=csv xlsx	
GET	/api/reports/expense/export?format=csv xlsx	

com.company.hrportal.admin

System settings, user management, audit logs, scheduler

Class	Type	Responsibility
AdminController	Controller	Settings, users, audit, scheduler APIs
UserManagementService	Service	Role change, activate, deactivate users
AuditLogService	Service	Insert-only audit log writes
SchedulerAdminService	Service	Spring Batch job trigger + status query
SystemSettingsService	Service	Key-value config read/write with cache
AuditLogRepository	Repository	audit_logs — insert only, no update/delete

API Endpoints

GET	/api/admin/settings	All system settings
PUT	/api/admin/settings/{key}	Update a setting
GET	/api/admin/users	User list with role
PUT	/api/admin/users/{id}/role	Change user role
PUT	/api/admin/users/{id}/status	Activate/deactivate
GET	/api/admin/audit	Audit log (paginated, filtered)
GET	/api/admin/scheduler	Job list + last run status
POST	/api/admin/scheduler/{job}/trigger	Manual trigger

com.company.hrportal.common

Shared utilities, file storage, exception handling

Class	Type	Responsibility
FileStorageService	Service	AWS S3 upload, download, pre-signed URL
GlobalExceptionHandler	ControllerAdvice	Standard error response format
ApiResponse	Model	Generic response wrapper { status, data, error }
PageableResponse	Model	Pagination wrapper
AuditableEntity	Base	@MappedSuperclass — created_at, updated_at
SecurityUtils	Utility	Get current user from SecurityContext
DateUtils	Utility	Working day calculation, date formatting

API Endpoints

No	direct APIs	shared by all modules
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Part 3 — Frontend Module Design

Vue 3 + TypeScript + Vuetify + Pinia · Module-per-feature structure under `src/modules/` · Each module owns its views, components, store, and service

src/modules/auth

6 files

File	Purpose
<code>views/LoginView.vue</code>	SAML redirect, local login form, error display
<code>views/SessionExpiredView.vue</code>	Timeout message + re-login CTA
<code>stores/authStore.ts</code>	Pinia — JWT storage, user state, role
<code>services/authService.ts</code>	Axios — <code>/auth/me</code> , <code>/auth/refresh</code> , <code>/auth/logout</code>
<code>guards/authGuard.ts</code>	Vue Router guard — checks JWT + role
<code>composables/useCurrentUser.ts</code>	Composable — exposes user, role, <code>hasRole()</code>

src/modules/employee

9 files

File	Purpose
<code>views/ProfileView.vue</code>	My profile — view + inline edit
<code>views/OnboardingView.vue</code>	Checklist with progress bar
<code>views/EmployeeDirectoryView.vue</code>	Search + filter table (HR/Manager)
<code>views/EmployeeDetailView.vue</code>	HR view — full profile edit
<code>views/AddEmployeeView.vue</code>	Create employee form
<code>components/ProfileCard.vue</code>	Reusable employee summary card
<code>components/DocumentUpload.vue</code>	S3 upload with progress
<code>stores/employeeStore.ts</code>	Pinia — employee list cache
<code>services/employeeService.ts</code>	Axios — all employee APIs

src/modules/leave

9 files

File	Purpose
<code>views/MyLeaveView.vue</code>	Balance cards + apply form + history

views/LeaveCalendarView.vue	Vuetify calendar + team leaves overlay
views/LeaveApprovalsView.vue	Manager — pending requests table
views/LeavePolicyView.vue	HR Admin — policy configuration
components/ApplyLeaveDialog.vue	Leave form modal with date picker
components/LeaveBalanceCard.vue	Per-type balance with visual progress
components/ApprovalActionPanel.vue	Approve/Reject with comment
stores/leaveStore.ts	Pinia — leave state
services/leaveService.ts	Axios — all leave APIs

src/modules/expense

8 files

File	Purpose
views/MyExpensesView.vue	Claims list + submit form
views/TeamExpenseApprovalView.vue	Manager approval queue
views/FinanceExpenseQueueView.vue	Finance payment queue
components/SubmitExpenseForm.vue	Multi-step: details + receipt upload
components/ReceiptUploader.vue	Drag-drop receipt upload with preview
components/ApprovalTimeline.vue	Stepper showing approval history
stores/expenseStore.ts	Pinia — expense state
services/expenseService.ts	Axios — all expense APIs

src/modules/payroll

7 files

File	Purpose
views/MyPayslipsView.vue	Payslip list + detail view
views/PayrollUploadView.vue	CSV upload + validation error table
views/PayrollPublishView.vue	Preview batch + publish confirmation
views/PayrollHistoryView.vue	Historical batches table
components/PayslipCard.vue	Gross / Deductions / Net summary
components/PayrollErrorTable.vue	Row-level CSV errors
services/payrollService.ts	Axios — all payroll APIs

src/modules/manager

5 files

File	Purpose
views/ManagerDashboardView.vue	Summary cards + quick action links
views/TeamOverviewView.vue	Team member list with status
views/OrgChartView.vue	D3 / Vue-based org tree
components/PendingBadge.vue	Approval count badge for nav
services/managerService.ts	Axios — dashboard, team, orgchart APIs

src/modules/reports

6 files

File	Purpose
views/HeadcountReportView.vue	Filter form + chart + table
views/LeaveReportView.vue	Leave utilisation chart
views/ExpenseReportView.vue	Expense summary with category bars
components/ReportFilterBar.vue	Shared filter component
components/ExportButton.vue	CSV/Excel download with loading state
services/reportService.ts	Axios — all report APIs

src/modules/admin

6 files

File	Purpose
views/SystemSettingsView.vue	Key-value settings table with inline edit
views/UserManagementView.vue	User list with role badges
views/AuditLogsView.vue	Filterable audit log table
views/SchedulerView.vue	Job list with last-run status + trigger
views/EmailTemplatesView.vue	Template list with rich text editor
services/adminService.ts	Axios — all admin APIs

src/router / stores / layouts / components

12 files

File	Purpose
router/index.ts	All routes with role-based meta
router/guards.ts	Auth guard + role guard
stores/uiStore.ts	Pinia — sidebar, loading, notifications badge
stores/notificationStore.ts	Pinia — in-app notifications + SSE
layouts/DefaultLayout.vue	Sidebar + topbar + router-view
layouts/AuthLayout.vue	Centered layout for login/expired pages
components/AppSidebar.vue	Role-aware navigation menu
components/AppTopBar.vue	User menu + notification bell
components/DataTable.vue	Reusable Vuetify DataTable wrapper
components/ConfirmDialog.vue	Generic confirm/cancel modal
services/apiClient.ts	Axios instance — JWT interceptor + 401 handler
services/sseService.ts	SSE connection management + reconnect logic